

Multi-Asset Time-Series Momentum Risk-aware portfolio construction using liquid ETF proxies

This report documents a pre-specified research process. Results are reported only after data validation, validation-period model selection, and a single untouched out-of-sample evaluation.

Executive summary

Report Do medium-term price trends across liquid asset classes support a diversified, implementable systematic portfolio after transaction costs?

Method Daily time-series-momentum signals across ten ETF proxies, tested first with equal absolute weights and then with inverse-volatility sizing.

Current status Data acquisition and integrity checks are complete. Signal, portfolio, validation, and out-of-sample results will be added sequentially.

Decision standard Evidence must remain credible across predefined parameter, cost, and rebalancing robustness checks; a high in-sample Sharpe ratio alone is not sufficient.

1 Research question and hypotheses

1.1 Primary question

Do medium-term price trends contain enough persistent information across liquid asset classes to construct a diversified systematic portfolio with superior risk-adjusted characteristics relative to reasonable passive benchmarks?

1.2 Hypotheses

H_0 . Past medium-term returns contain no economically useful information sufficient to produce a robust portfolio after realistic implementation assumptions.

H_1 . Positive medium-term trends tend to persist and negative medium-term trends tend to continue sufficiently to support a diversified systematic trend-following portfolio.

Secondary hypothesis. Volatility-aware sizing improves the risk-adjusted characteristics and robustness of the raw momentum strategy.

2 Data

2.1 Universe and source

The analysis uses adjusted daily ETF prices obtained from Yahoo Finance via `yfinance`. The frozen universe covers equities, government rates, credit, real assets, and currency exposure. The aligned sample contains 4,780 common trading dates from 2 July 2007 through 1 July 2026.

Table 1: Frozen research universe

@ l X@	Ticker	Risk source	Role in the universe
	SPY	US equities	Broad US large-cap equity beta
EFA	Developed ex-US equities	International developed-market equity exposure	
EEM	Emerging equities	Higher-volatility emerging-market equity risk	
IEF	Intermediate Treasuries	Moderate-duration US government-bond exposure	
TLT	Long Treasuries	Long-duration US government-bond exposure	
	LQD	Investment-grade credit	Corporate credit-spread risk
	GLD	Gold	Precious-metal and real-rate-sensitive exposure
DBC	Broad commodities	Diversified commodity-futures exposure	
	UUP	US dollar	Long USD currency exposure
	VNQ	US real estate	Listed US real-estate exposure
Yahoo Finance, retrieved 10 August 2026.			

2.2 Data policy and limitations

Prices are adjusted for splits and distributions. The analysis retains only dates common to all ten instruments; missing values are not forward-filled. The use of ETF proxies, Yahoo Finance data, and a strict shared-history sample are material limitations and remain explicit throughout the report.

[figures/correlation_matrix_heatmap.svg](#) *Daily – return correlation matrix of the frozen ETF universe*

3 Signal definition

For asset i at date t and lookback L , momentum is defined as

$$M_{i,t,L} = \frac{P_{i,t}}{P_{i,t-L}} - 1.$$

The signed signal is

$$s_{i,t} = \{ +1 \text{ if } M_{i,t,L} > 0, 0 \text{ if } M_{i,t,L} = 0 \text{ or history is insufficient, } -1 \text{ if } M_{i,t,L} < 0.$$

The baseline uses $L = 252$ trading days. Candidate lookbacks are selected only using the validation protocol described below. Targets formed with close data at t are applied by the backtesting engine to the return ending at $t + 1$.

4 Portfolio construction and implementation

4.1 Equal-weight momentum baseline

The baseline target weight is $w_{i,t} = s_{i,t}/10$, rebalanced monthly at 100% gross exposure. Net exposure is signal-dependent and is not constrained to zero.

4.2 Inverse-volatility comparison

For a trailing volatility window W ,

$$\hat{\sigma}_{i,t} = \text{std}(r_{i,t-W+1:t})\sqrt{252}, \quad w_{i,t} \propto \frac{s_{i,t}}{\hat{\sigma}_{i,t}}.$$

Weights are normalised to 100% gross exposure, subject to a 35% individual absolute-weight cap. No portfolio-level volatility target or leverage is used in the initial study.

4.3 Costs and benchmarks

Transaction costs are modelled as one-way proportional costs per unit of turnover. The baseline is 5 bp per asset, with sensitivity at 0, 1, 2, 3, 5, 10, and 20 bp. Benchmarks are the equal-weight long-only universe, SPY buy-and-hold, and individual buy-and-hold asset results.

5 Research design

Table 2: Temporal separation and decision protocol

Period	Dates	Purpose
Train	2008-07-01 to 2015-06-30	Infrastructure, diagnostics, and development
Validation	2015-07-01 to 2019-06-30	Predefined model comparison and selection
Test	2019-07-01 to 2026-07-01	Untouched out-of-sample evaluation

The validation procedure is staged. First, predefined momentum lookbacks are compared under equal-weight monthly construction. Second, the selected lookback is held fixed while equal-weight and inverse-volatility variants are compared. Rebalance frequency is a robustness analysis rather than a parameter chosen solely by maximum Sharpe ratio. No model decision is revised in response to test-period performance.

6 Results

Populate this section only after the predefined validation process has completed. Keep equal-weight and inverse-volatility results side-by-side, including results that do not support the hypothesis.

Table 3: Performance summary — to be populated

Portfolio	CAGR	Volatility	Sharpe	Max DD	Turnover
Equal-weight momentum					
Inverse-volatility momentum					
Equal-weight long-only					
SPY buy-and-hold					

figures/strategy_vs_benchmarks.svg *Net equity curves for strategy variants and passive benchmarks.*

7 Interpretation, limitations, and conclusion

This section will interpret results conservatively, including periods of failure, attribution by asset and regime, robustness outcomes, and the limits imposed by ETF proxies, simplified execution, transaction-cost assumptions, and parameter-selection uncertainty.